Toward Symbolic-Aware Music Language Models

Compact Learned Symbolic Planning Tokens for Controlled Music Fusion

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Audio language models handle sound well — but musical structure lives in MIDI, not in waveforms. **Can we inject symbolic structure into a LALM in a controlled, verifiable way?**

Audio Tokens Do Not Reliably Encode Musical Structure

What audio tokens can encode	**What they miss**
Timbre, loudness, texture	Key signature, chord progression
General instrument colour	Stem-level instrument identity
Rough tempo feel	Exact meter and beat phase
High-level scene	Phrase boundaries, section structure

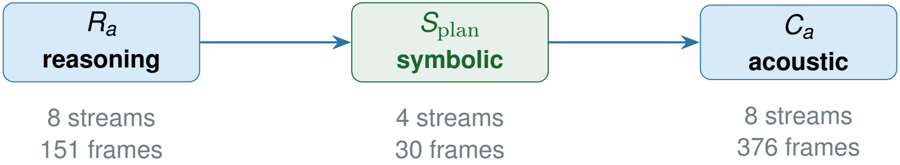
Verifying that symbolic injection is *useful* requires more than comparing to no-injection: a shuffled or dummy stream of the same length could produce the same gain from capacity alone. **Our whole evaluation is designed around this distinction.**

Three Research Questions Organize This Work

Question	What we need to show
RQ1	
Acoustic utility	Aligned symbolic stream reduces acoustic token CE more than a shuffled or dummy stream of the same size.
RQ2	
Alignment specificity	The gain comes from *alignment-specific content*, not token budget. Requires aligned $>$ shuffled AND aligned $>$ dummy.
RQ3	
Structural content	Symbolic stream encodes music-structural attributes recoverable by probing, verified against a zero-feature baseline.

Each comparison uses the same aligned / shuffled / dummy protocol, so gains can be attributed to symbolic content rather than extra tokens alone.

Our Approach: S_{plan} as a Third Stream in UniAudio 2.0



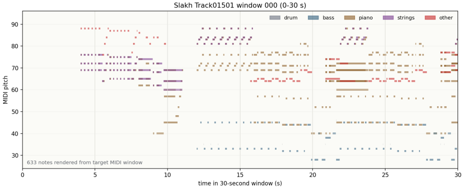
S_{plan} uses a 4-stage RVQ codec trained on MIDI-derived frame features (pitch-class, instrument, meter, density). Total sequence: $151 + 30 + 376 = \mathbf{557}$ tokens — **0% truncation** under 1,024 limit.

Raw MIDI appending fails: 2,000–8,000 tokens per window saturates the context, and variable-length sequences cannot be shuffled or dummy-controlled uniformly. S_{plan}'s fixed-length RVQ design makes the evaluation protocol possible.

Dataset: Slakh2100 — Synthesized Paired Audio-MIDI

Split	Tracks	Windows
Train	1,289	11,279
Validation	270	2,334
Test	151	1,385

- Synthesized with professional-grade sample-based virtual instruments
- Track-level split *before* windowing — no phrase contamination
- Each window: 30 s audio + ground-truth aligned MIDI
- All CE results in this talk are on the held-out **test** set



Example 30 s piano roll — Slakh track 01501

Method: MIDI Feature Extraction — 53-Dim Frame Vector

Extracted at 100 ms resolution from ground-truth MIDI

Feature group	Dims	Content
Binary features	41	Instrument-class presence (34 Slakh classes), pitch-class activation (12), beat/downbeat phase
Continuous features	12	Instantaneous tempo (BPM), note density (events/s), polyphony count, energy proxy
Frame vector	**53**	Normalized per-dimension; $\approx$300 frames per 30 s window

MIDI features are extracted from the *ground-truth* MIDI file, not transcribed from audio. The 300-frame sequence is compressed by the codec encoder to a fixed **30 frames** — one frame per second of audio.

Method: S_{plan} RVQ Codec Architecture

Encoder	2-layer 1D CNN + temporal pooling; maps variable-length input to 30 frames
Codebook	4 RVQ stages, 512 codes each; EMA updates + codebook restart
Training loss	BCE for binary dims + MSE for continuous dims; 42,000 steps

$4 \times 30 = 120$ tokens per window. Same RVQ format as C_a — stream-native positional encoding.

Experiment Setup: LoRA Fine-tuning with Learned Gate

Component	**Configuration**
Base model	UniAudio 2.0 (frozen weights)
Adapters	LoRA on 140 attention modules
Trainable params	$\sim$10.8M (reason-only) $\sim$253M (symbolic variants)
Symbolic gate	Learned scalar; controls S_{plan} influence on C_a prediction
Hardened four-condition comparison (primary result)	
Steps / batch	800 steps; batch 1 + gradient accumulation 16
Learning rates	LoRA 10^{-5} / symbolic 3×10^{-4} / gate 2×10^{-3}
Seed	Fixed; all four conditions identical except stream variant

Gate ≈ 0: model ignores S_{plan}. Gate > 0: symbolic stream admitted to C_a prediction. Primary result: aligned 0.177 $>$ shuffled 0.167 $>$ dummy 0.163.

Evaluation Protocol: Every Comparison Uses All Four Conditions

Condition	Content of S_{plan} slot	What a gain here proves
Aligned S_{plan}	Tokens from the *same* window's MIDI	Symbolic content + temporal alignment
Shuffled S_{plan}	Tokens from a *different* window's MIDI	Token statistics intact; alignment broken
Dummy stream	Zero / random RVQ codes	Token-budget / capacity effect only
Reason-only	No S_{plan} slot at all	Base model without any symbolic stream

Evaluation criterion: aligned must beat *both* shuffled and dummy. Shuffled preserves statistics but breaks alignment; dummy removes all content. Together they isolate the alignment-specific effect from pure capacity.

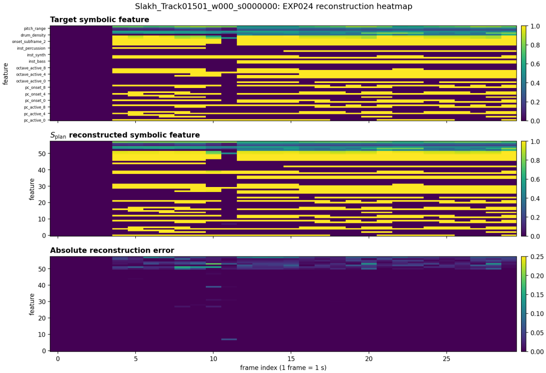
Prerequisite: The Symbolic Codec Reconstructs Faithfully

Metric	Value
Validation reconstruction loss	0.00918
Binary feature accuracy	**0.99777**
Continuous feature MSE	**0.00172**
Codebook utilization — books 1/2/3/4	31% / 72% / 81% / 81%
Fixed window `w000` binary acc / MSE	1.000 / 0.00174
Fixed window `w001` binary acc / MSE	1.000 / 0.00246
Fixed window `w003` binary acc / MSE	1.000 / 0.00130

Books 2–4 reach 72–81% utilization — no codebook collapse. This validates the codec before downstream comparison: later differences are not explained by reconstruction failure.

Prerequisite: Feature Heatmap — Target vs. Reconstruction

Slakh Track01501, window 0 (0–30 s) — binary acc 1.000, MSE 0.00174



Rows: feature dimensions (binary instrument flags, pitch-class bits, continuous values). Columns: 30 time frames (1 s each). Left = ground-truth MIDI features. Right = S_{plan} reconstruction.

RQ1 — First Positive Signal: Proxy Prediction Task

Frozen codec + proxy head; 5,000 steps, 1,385 test windows

Condition	Valid CE	Test CE	Test PPL
Reason only	7.565	7.657	2116
Low-rate rule summary	7.553	7.645	2089
Aligned S_{plan}	**7.507**	**7.596**	**1991**
$\hookrightarrow$ Shuffled S_{plan}	7.538	7.631	2061
$\hookrightarrow$ Dummy stream	7.602	7.694	2196
$\hookrightarrow$ Random stream	7.540	7.632	2064

First evidence for RQ1. Aligned beats both corrupted controls. Temporal alignment of symbolic content to its own audio window drives the gain. This proxy task is followed by end-to-end integration.

RQ1 — Replication: Stream-Native Interface Confirms the Gain

Full $R_a \to S_{\text{plan}} \to C_a$ layout with matched positional encoding

Condition	Test CE	Test Acc	Gate
Aligned S_{plan}	**5.766**	**0.1097**	0.003
$\hookrightarrow$ Shuffled	5.785	0.1067	0.003
$\hookrightarrow$ Dummy	5.785	0.1068	0.003

RQ1 holds with end-to-end integration. Replicates as a native third stream, not a side-channel. The effect also appears outside the proxy setup. Sequence 557 tokens, **0% truncation**.

RQ1 — Primary Result: Hardened Four-Condition Comparison

All four conditions identical except stream variant — matched hyperparameters, fixed seed

Condition	Test CE	Test PPL	Acc	Gate
Reason-only baseline	5.859	350	0.104	0.000
Aligned S_{plan}	**5.600**	**270**	**0.121**	**0.177**
$\hookrightarrow$ Shuffled S_{plan}	5.614	274	0.119	0.167
$\hookrightarrow$ Dummy stream	5.614	274	0.119	0.163

RQ1 answered positively — most rigorous evidence. Gate confirms the model opens the symbolic slot more for aligned (0.177) than shuffled (0.167) or dummy (0.163).
Scope: oracle MIDI input; semantic-token CE only.

RQ2: The Gain Is Alignment-Specific, Not Token Capacity

Effect	**Δ CE**
Stream capacity (any symbolic stream vs. reason-only)	-0.245
Alignment-specific (aligned vs. both controls)	-0.014
Total (aligned vs. reason-only)	**-0.259**

Comparison	**Steps**	**Aligned vs. shuffled**
Proxy task	5,000	-0.034
Stream-native warmup	3,000	-0.018
Hardened comparison	800	-0.014

RQ2 answered positively. The direction is consistent across independent comparisons. Corroborated by gate: aligned 0.177 $>$ shuffled 0.167 $>$ dummy 0.163.

RQ3: Music Structure Is Recoverable from S_{plan}

MIR attribute	Aligned S_{plan}	Zero-feature control
Weak key accuracy	**0.557**	0.126
Pitch-class macro-F1	**0.854**	0.830
Instrument macro-F1	**0.501**	0.426
Segment density acc.	0.874	0.879
Instrument micro-F1	0.791	0.835
Meter accuracy	0.783	0.855

The zero-feature baseline separates structural signal from dataset priors. Weak key (+0.431 gap) and pitch-class (+0.024 gap) are the strongest axes. **RQ3: qualified positive answer** — key and pitch-class confirmed, meter and instrument micro-F1 excluded.

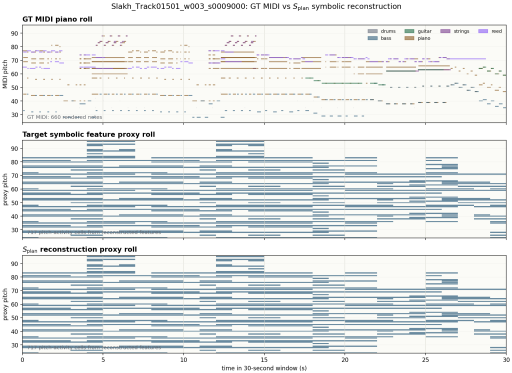
RQ3: Which Axes Are Supported?

Axis	Gap	Status
Weak key accuracy	$+0.431$	**Strongest support.**
Pitch-class macro-F1	$+0.024$	Positive feature-specific signal.
Instrument macro-F1	$+0.075$	Positive, with class-imbalance sensitivity.
Instrument micro-F1	-0.044	Prior-dominated. Common-class baseline wins.
Meter accuracy	-0.072	Prior-dominated. Common-meter baseline wins.

Report each MIR axis with its zero-feature gap; absolute accuracy alone is not enough.

RQ3: Qualitative — Reconstruction vs. Ground-Truth Piano Roll

Slakh Track01501, window 3 (90–120 s) — binary acc 1.000, MSE 0.00130



Left: ground-truth MIDI piano roll. Right: pitch-region reconstruction from the S_{plan} codec.

This shows symbolic feature reconstruction, not MIDI transcription or waveform generation.

Design Constraint: Stream-Native Interface Is Necessary

Side-channel insertion:

Condition	Test CE	vs. reason-only
Reason-only	**5.360**	baseline
Aligned S_{plan}	5.511	$+0.151$
Shuffled S_{plan}	5.512	$+0.152$
Dummy	5.510	$+0.150$

Same root cause in both cases: symbolic tokens must enter through the same stream-native positional encoding as R_a and C_a. **This is why the RQ1/RQ2 positive results are non-trivial.**

Current Scope and Next Steps

Gap	Why it matters
No decoded generation	CE results are on semantic tokens only. No waveform decoded from aligned vs. shuffled.
Oracle MIDI input	Aligned MIDI unavailable at inference on real audio. Needs audio-to-S_{plan} predictor.
No LLM-supervised tokenizer	Text-language grounding (open QA, captioning) not yet trained.
MIR controls in progress	Label-shuffle and target-prior controls will strengthen the probing analysis.
Slakh only	No real-recording or cross-domain transfer evidence.

Next priority: decode C_a from the aligned checkpoint to waveform and compare perceptually against shuffled — connecting the token-level result to perceptual audio evidence.

Conclusion: Three Research Questions Answered

RQ	Finding
RQ1	**Acoustic utility — Positive.** Aligned S_{plan} reduces CE from 5.859 to 5.600; consistent across all three comparisons.
RQ2	**Alignment specificity — Positive.** Aligned $>$ shuffled $>$ dummy ($\Delta \in [-0.034, -0.014]$); gate $0.177 > 0.167 > 0.163$.
RQ3	**Structural content — Qualified positive.** Weak key ($+0.431$) and pitch-class ($+0.024$) confirmed; meter and micro-F1 excluded.

Design constraint (interface failures) makes RQ1/RQ2 non-trivial. Multi-condition protocol (aligned/shuffled/dummy + zero-feature probing) is the methodological contribution.

Backup: Per-Codebook Cross-Entropy

Symbolic benefit distributed across all 8 acoustic codebooks (hardened comparison)

Condition	**cb0**	**cb1**	**cb2**	**cb3**	**cb4**	**cb5**	**cb6**	**cb7**
Reason-only	5.60	6.00	3.45	5.28	6.02	6.48	6.87	7.16
Aligned S_{plan}	**5.24**	**5.71**	**3.19**	**4.98**	**5.75**	**6.25**	**6.68**	**7.01**
Shuffled	5.26	5.73	3.20	4.99	5.76	6.26	6.69	7.02
Dummy	5.26	5.73	3.20	4.99	5.76	6.26	6.69	7.02

Symbolic benefit is distributed across all 8 codebooks. Books 0, 2, and 3 show the largest gains, suggesting symbolic structure acts as a long-range prior that helps coarser-grained acoustic prediction first.

Backup: Per-Window Case Spectrum (Stream-Native Warmup)

Mixed per-window margins in the stream-native warmup

Window	Aligned CE	Best control	Margin	Note
`w000`	5.226	5.280	$+0.054$	Strong positive
`w001`	6.159	6.163	$+0.004$	Small positive
`w002`	6.097	6.091	-0.006	Slight negative
`w003`	5.876	5.869	-0.007	Slight negative

Aggregate positive result holds even though individual windows show mixed margins. The warmup contains both strong positive and near-zero examples.